Large Data Set: OCR A

Questions 1–13

Question 1. State the following facts about the Large Data Set:

a) the 10 different categories of method of travel to work (excluding 'other').
b) the extra category to ensure that all people are included in the travel data collection.
c) the 2 different years from which the data was collected.

Question 2. What happened to the authorities between 2001 and 2011 which has an impact when comparing the two years with this data?

Question 3. Kyra states that if she wants to find all the people who come to work by park and ride the only category she needs to consider is "bus, minibus or coach". Explain why Kyra is wrong.

Question 4. In 2011, Abdullah was unemployed the week before the census, but started a new job close to his house the day he filled it in. Which category should he have identified as?

Question 5. Toby is considering the 2011 age data by authority. He notices that Exeter has 14155 people aged 20-24 whilst Ashford has 6245 aged 20-24 despite a similar total number of residents. Explain a reason why this might be the case.

Question 6. Jakub states that in 2001, the number of people using the Underground, metro, light rail and tram in the South East and the South West can be summarised using the following statistics:

$$\sum x = 1915, \quad \sum x^2 = 196\,261, \text{ and } n = 36$$

a) Calculate the mean and the standard deviation, where $s_x = \sqrt{\dfrac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$, giving your answer to 1 decimal place.

Jakub's teacher explains he is wrong as he has missed a data point.

b) Calculate the new mean and standard deviation which includes this data point.

Question 7. State how a baby which is less than 1 year is categorised in the Large Data Set.

Question 8. For each of the following, given a sample of number of cyclists per authority is being taken from the Large Data Set, write down the name of the sampling method used.

a) The order of the authorities is randomised, then every 100th is noted.
b) The order of the authorities is randomised, then the first 5 from each region is noted to obtain 50 authorities.
c) The data set is sorted by authorities and the proportion of each authority is calculated. Using a random number generator and the position in the spreadsheet, each authority is selected to the same proportion as the full data set.
d) The order of the authorities is randomised, then the first 20 authorities are chosen.

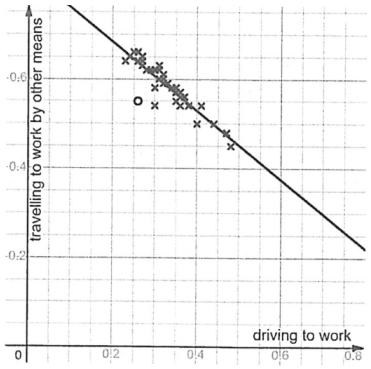
Question 9. The table below summarises some age data about London.

	2001	2011
Age 0 to 4	478187	591495
Age 5 to 14	887190	939674
Age 15 to 24	947810	1101631
Age 25 to 44	2533089	2903920
Age 45 to 64	1434225	1732472
Age 65 to 74	468067	473058
Age 75 to 84	310553	308661
Age 85 to 94	112970	123030
	7172091	8173941

a) Is there a significant increase in the proportion of 25–44-year-olds living in London?
b) By comparing the number of 15-24-year-olds living in London in 2011, to the 5-14-year-olds living in London in 2001, comment on the change in this demographic.
c) Explain why the final category includes data outside of the category range.

Question 10. The scatter graph below shows the proportion of working people "travelling to work by any means which is not driving" against people "driving a car" in the North West in 2001.
The product moment correlation coefficient, r, is -0.899 to 3 significant figures.

a) One of the points is identified with a circle rather than a cross.
By considering this point, explain why all the points do not lie exactly in a line with $r = 1$.



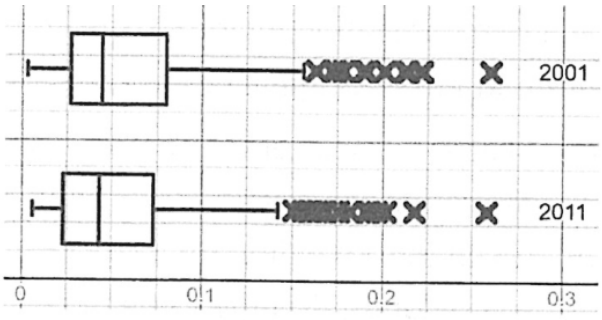
The least squares regression line is given by

$$y = -0.7718x + 0.8374$$

where y is the proportion of people not travelling to work by driving and x is the proportion driving to work.

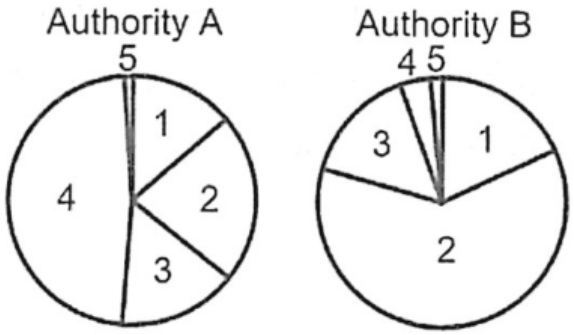
b) By using the regression line, calculate how many of the people in the North West would typically drive to work if they were in a local authority where 50% of people drive to work.

Question 11. The proportion of people travelling by bus, minibus or coach in 2001 and 2011 is shown in the boxplots below.



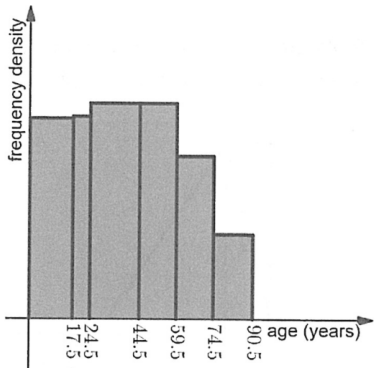
Compare the two data sets.

Question 12. Consider the two pie charts below showing data from 2001. One is in London and the other is in a rural location.



The categories are 'any form of public transport', 'working from home', by 'bicycle or on foot', 'car as a passenger or a driver' and 'other' but not in that order.
Identify which is which and make suggestions on which categories 1-5 will be, given they represent the same type of data in both pie charts.

Question 13. Prerequisite knowledge for Q13: Normal Distribution Alex creates a histogram to show the distribution of the age of people in the South East in 2011.
There are 8 634 750 people living in this region in 2011.



The heights of the bars are $\{213, 215, 229, 229, 173, 90\}$

a) Estimate the mean and the standard deviation of the data.
b) Estimate the number of people aged 68 or above. Explain why this is an estimation.
c) Would it be reasonable to assume that this data is normally distributed? Fully justify your answer.
d) Explain how the histogram created by Alex could be improved to better represent the data.